

Measuring Effects of Spatial Visualization and Domain on Visualization Task Performance: A Comparative Study

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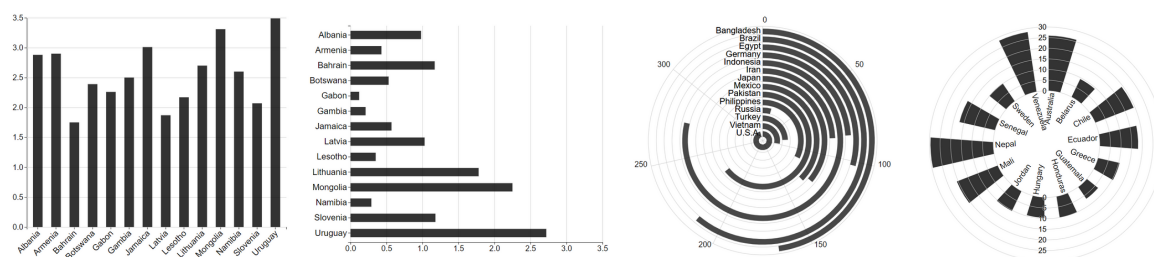


Fig. 1: Stimuli created for this study designed to elicit differences in spatial visualization. Stimuli display a bar chart rotated along Cartesian and radial axes: Vertical, Horizontal, Radial, and Circular Bar Plot.

Abstract—Understanding one's audience is foundational to creating high impact visualization designs. However, individual differences and cognitive abilities influence interactions with information visualization. Different user needs and abilities suggest that an individual's background could influence cognitive performance and interactions with visuals in a systematic way. This study builds on current research in domain-specific visualization and cognition to address if domain and spatial visualization ability combine to affect performance on information visualization tasks. We measure spatial visualization and visual task performance between those with tertiary education and professional profile in business, law & political science, and math & computer science. We conducted an online study with 90 participants using an established psychometric test to assess spatial visualization ability, and bar chart layouts rotated along Cartesian and polar coordinates to assess performance on spatially rotated data. Accuracy and response times varied with domain across chart types and task difficulty. We found that accuracy and time correlate with spatial visualization level, and education in math & computer science can indicate higher spatial visualization. Additionally, we found that motivational differences between domains could contribute to increased levels of accuracy. Our findings indicate discipline not only affects user needs and interactions with data visualization, but also cognitive traits. Our results can advance inclusive practices in visualization design and add to knowledge in domain-specific visual research that can empower designers across disciplines to create effective visualizations.

Index Terms—Human-subjects quantitative studies, visualization, perception, bar chart, discipline, domain-specific, education, empirical evaluation, spatial ability, cognitive abilities

1 INTRODUCTION

Data shapes modern society and affects every aspect of our lives: political, social, and physical. Information visualization is just as prevalent – it lives in magazines, news outlets, scientific papers, blogs, and countless online platforms to facilitate engagement with data in a way text alone cannot. The data visualization community has long explored how to design universally accessible and engaging visualizations that empower communities to understand and reason with data [67]. Information visualization is not simply a matter of reading data properties encoded in visual form – there are complex cognitive activities at work that are influenced by visual structure [94]. Visualization research has followed several paths toward understanding users and their needs including understanding how cognitive abilities affect perception and how disciplines differ in terms of data and tasks [53, 65, 74, 84, 94]. Education psychology has explored the connections between education background, performance, and cognitive spatial abilities, noting these are not easily separated in individuals [7, 87]. Recently, work has been done in visualization toward integrating these tracks of research to move away from universal guidelines for the monolithic “user” and towards effective design that considers the differing cognitive traits of target disciplines [41].

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We build on this past research to explore and investigate the cognitive trait of spatial visualization (SV) (i.e., abilities involving retention, manipulation, and rotation of visual images [58]) and its interaction within and across not only professional domains but also educational backgrounds. We chose to center this study around SV due to its history of study within information visualization and educational and professional domains; SV is a measurable cognitive ability that not only affects interpretation of visual information, but it differs among domains [7, 50, 65, 75, 84, 87, 94]. However, there are gaps in bridging research in SV, domain, and information visualization. The aim of our study is to increase impact of visualization design by taking into account cognitive interactions between domain experiences and SV, as neither alone can account for differences in visual interaction.

We focus our study on the disciplines of business, law & political science, and math & computer science; the use of visualization is core to these disciplines, from decision making to policy development to data analysis. In this paper, we present past research related to SV in psychology and visualization design that led us to create hypotheses around how our target domains would perform SV assessments and relevant visual tasks. We pose a study methodology that links SV to visual task paradigms that evoke the same cognitive functions. We offer a detailed statistical analysis demonstrating that visualization task performance (accuracy and response times) varies with SV and discipline. Finally, we present how domain motivation differences concerning data visualization might affect performance on visual tasks.

Our study is the first to tie domain motivation differences and SV together to explain performance on prevalent visual stimuli and tasks resulting in novel design implications. Our results add to evidence that SV and domain interact to affect visual task performance and should

be considered as combined factors that can influence design choices; visualization design should consider not only the needs of varying domains, but the abilities of individuals in those domains [41]. This paper contributes toward cataloguing visualization performance differences amongst SV levels and domains that can empower individuals across disciplines to make memorable references and inferences effectively and efficiently using visualizations designed for them.

2 RELATED WORK

Our study builds on research in SV, education psychology, and information visualization. It bridges visualization research in user cognition and domain needs/expertise to inform design as a holistic process. We draw on methods from psychology and visualization to investigate the relationships between SV, domain, and visualization comprehension.

2.1 Spatial Visualization

Spatial ability is a cognitive ability referring to skills involved with retrieval, retention, and transformation of visual information [51, 58]. It impacts a viewer's use of graphics as it influences the processing of spatial relationships between graphical elements and/or their cognitive resource allocation [83]. SV is a specific aspect of spatial abilities that involves the ability to remain unconfused by varying orientations or rotational positions in which a spatial pattern may be presented, thus it is elicited most in tests that involve manipulation or transformation of a visual stimulus [51, 58]. SV allows an individual to compare different encodings of visualizations quickly and accurately [58], and has been connected to spatial scaling, figure construction, magnitude sense, numerical estimation, subdivisions of charts, and even textual spatial analogies [61, 92]. In all, high SV implies the ability to manipulate and change internal representations of visualizations, leading to measurable outcomes when interacting with external visualizations [21, 44, 55].

2.2 Education Psychology and Spatial Visualization

Spatial abilities, including SV, have often been tested in educational settings as an indicator of increased performance in STEM subjects, especially in mathematics [7, 33, 47, 61, 72, 75, 92]. Increased SV has also been tied to increased performance in students of geology and architecture [64, 72]. Additionally, substantial SV deficits have been identified as the primary component in certain learning disorders wherein students struggle to use visual representations of data [13].

Further research has been done into SV of students and professionals across a range of domains [16, 75, 87]. Burnett and Lane [16] tested college students in their first semester and again after two years of study in various subjects. They found that SV performance improved with advance in education across all subjects, but by a lower magnitude for humanities and social science students than math and physical science students. Shea et al. [75] and Wai et al. [87] conducted comprehensive studies assessing spatial abilities across domains, levels of education, and occupation. Both found students of humanities, education, law, business, and medicine to have lower spatial abilities than math/computer science, physical science, and engineering students. While these studies assessed spatial abilities of domains, they did not extend beyond implications for education. However, they demonstrate that specific domains and duration of experience influence spatial abilities, which in turn could influence individual ability to use information visualization. Evidence that it is difficult to separate SV abilities from education and professional choices could empower designers to create impactful visuals for specific communities.

2.3 Spatial Visualization in Information Visualization

A rich body of literature exists investigating spatial abilities in the context of information visualization as it is a key cognitive factor in interpreting visuals [41, 50, 65, 83–85, 88, 94]. Research has demonstrated that high SV, specifically, can correlate with higher recall, understanding, and increased task performance using data visualization [50, 65].

Kim et al. [50] observed that for both parallel coordinate plots and written descriptions of data, participants with high SV had higher accuracy when recalling values. Vicente et al. [85] found low spatial ability corresponded with inferior performance on retrieval tasks within

visual file structures: low spatial individuals were two times slower and more likely to get lost in the structure than high spatial individuals. Kellen [49] and Ottley et al. [65] found that high spatial individuals had significantly increased accuracy in their performance on solving conditional probability problems when aided by a visualization. On their study using line-up tests, VanderPlas and Hofmann [83] found that performance of undergraduate students correlated with visual ability and whether a student was a STEM major. Studies also indicated that speed is often related to understanding [78, 84] and was sometimes the bigger indicator of lower cognitive abilities compared to accuracy [25, 88]. Wenhong [88] found that high spatial ability led to reduced response times and higher accuracy with graphics and table visualizations, and interestingly that those with low spatial ability still chose graphic visualizations for tasks. Wenhong noted that often individuals are not aware of their own cognitive style, but still prefer visual representations of data. These findings indicate space for accessible design in information visualization for low spatial individuals [90].

While cognitive abilities are thoroughly studied in visualization design, educational or professional domain is often overlooked as a relevant influence and not reported on [49, 50, 65, 84, 85] or referenced, but not analyzed [25, 88]. However, there is also rich research in domain-focused participatory methodologies that imply domain impacts design.

2.4 Designing Visualization for Domain

Literature on visualization design highlights interaction with domains to inform the design process, but they do not explicitly consider differing cognitive traits of the domains [27, 40, 53, 62, 73, 74, 77, 91]. The design study methodology proposed in [74] emphasizes the fundamental need to design with and for specific domains. Distinct from discovering practices and needs of those domains, it also advocates for designers to be aware of literature pertaining to relevant design problems, but not specifically to cognitive and perceptive knowledge, which are themselves distinct from visual task needs of domain experts. Other methods like participatory design, design by immersion, and domain liaisons [40, 77] potentially have benefits due to implicitly integrating design reflective of cognitive capabilities, without the need to explicitly rely on any empirical, systematic analysis of cognitive abilities.

Kirby & Meyer [53] and Munzner [62] further emphasize collaboration with domain experts and employment of applied psychology methodologies to measure cognition and perception, but do not touch upon capitalizing on the cognitive features of domains in guiding design. Peck et al.'s multidimensional model of individual differences in HCI [68] separates individual experiences/bias from cognitive abilities in visualization design; similarly, Xiong et al. [91] and Dasgupta et al. [27] focus on domain knowledge biasing interaction with visuals, while not referencing development of cognitive abilities as a factor in performance. Treating an individual's background as orthogonal to their cognitive traits implies the two do not influence one another, which could lead to missed opportunities and ineffective design choices.

Hall et al. [41] made an initial exploration into how SV differs between professional domains (Education, Chemistry, and Computer Science) and relates to visualization processing. They confirmed SV differences amongst their chosen disciplines and that domain and SV together influenced performance and qualitative strategies on tasks related to 2D to 3D representations of isocontour plots and scatter plots. Their findings imply differences in task performance and SV between domains and that these differences will endure with design implications for visualization. While Hall et al. combine cognitive abilities and domain, our work extends theirs to include a systematic exploration into design through a controlled study with globally diverse balanced samples of domains, inclusion of education background, evaluation of domain motivations concerning data visualization, and stimuli and data involving prevalent visual tasks. Research demonstrates there are ties between SV, domain, and visualization performance – we endeavor to further findings into how cognitive differences between domains relate to visualization performance to affect design decisions and empower designers to create more relevant and cognitively appropriate visuals.



Fig. 2: Activity sequence of the study. In Part 1, the paper is folded and punched above the line. Participants are given four multiple-choice responses, the selected choice is highlighted and participants can move forward. Scores are the number answered correctly out of 10 in 3 minutes.

3 MOTIVATION & HYPOTHESIS

Drawing on the outcomes and identified gaps from previous work, we sought out to study SV, motivational differences across domains, and their combined influences on visualization interaction. We ensured alignment between profession and education background as research demonstrates increased SV depending on education domain [16]. If occupation does not align with educational background, the abilities gained in education should be considered.

We chose to study individuals with professional profile and education background in Law and Political Science (LPS), Business, and Math and Computer Science (MCS). Differences between MCS and other disciplines have implications on visualization research findings as they are often “standard participants” for many studies on task performance and design choices; a considerable amount of information visualization is also created and studied by those in MCS related fields [19, 20, 83]. We chose Business as there is a wealth of literature exploring data visualization for business intelligence and business decisions, which often cites design and cognition research [34, 56, 81, 93]. However, we are unaware of any comparative studies between MCS and Business domain cognitive abilities applied to data visualization task performance. We chose LPS due to the relative lack of research investigating data visualization and law & politics, despite the imperative use of data in law & politics [45, 95]. However, given many governments have scientific advisory and consulting committees, and politicians make health and governance related decisions based on this advice [24], it may be worth exploring any differences in visual cognition between those creating and consuming visualizations for political decisions. In line with visualization domain research, we believe these domains’ differing experience around data visualization will bias and affect performance on visual tasks in a measurable way [27, 53, 74]. To gain a measurable understanding of how domain background affects interaction with visualization we included an assessment on motivational differences surrounding data visualization for analysis.

Additionally, these domains display varying levels of SV ability with MCS exhibiting higher than average abilities, and Business and LPS below average [87]. Given that SV affects interactions with visualization, there may be measurable differences in how these domains interact with data visualizations given their discipline. In order to claim differences in performance and SV level correlate with domain, we ensured a balanced sample from each domain for comparable results. Our hypotheses build upon existing research to demonstrate our methodology results in analogous findings and novel design recommendations for new domains. We aim to confirm if SV ability differs across domain, how SV level affects interactions with data visualization, and explore if domain background interacts with SV to affect performance in a quantitatively significant way. We explore these questions by investigating the following hypotheses.

H1: SV will differ across domains. Research has demonstrated systematic differences in SV abilities between domains [75, 87]. Drawing from these findings, we anticipate MCS will have the highest SV ability and LPS will have the lowest. As SV affects visualization interaction, these systematic differences between domains may impact visual reasoning in a discriminating way.

H2: Performance (accuracy and response time) will correlate with SV level. Prior work in information visualization shows accuracy and timing of visualization tasks vary with SV abilities [41, 50, 65, 83–85, 88, 94]. High spatial individuals tend to have higher accuracy and reduced response times. We expect to see comparable results, emphasizing the role of SV in performance

and use of visualizations.

H3: Performance (accuracy and response time) will correlate with domain such that average performance will vary between disciplines according to their level of SV. Hall et al. [41] found correlations between SV and task performance across Chemistry, CS, and Education for 3D visuals, implying that SV is an important underlying source of variations alongside domain experience. Specifically, we expect MCS to perform the best and LPS the worst following their SV levels.

H4: Performance (accuracy and response time) will correlate with domain such that average performance will vary between disciplines due to motivational differences. Visualization research demonstrates that needs, knowledge, bias, and experience differ by domain and that these factors can affect performance of tasks and visualization design [27, 40, 53, 74, 77, 91]. We anticipate marked differences in performance and motivations around data visualization between MCS, Business, and LPS given the three domains varying levels of interaction, creation, and consumption of data visualization (i.e., domain experience) [36, 93, 95].

4 METHODOLOGY

A two-part online study was created to address the research questions and hypotheses (see Fig. 2). Part 1 of the study consisted of a brief SV psychometric test, known as a paper folding or punch test. Part 2 consisted of stimuli and questions asking individuals to draw conclusions from visualizations. A short motivation assessment followed Part 2. Inspired by current methodology in vision science [32], we began by evaluating paradigms that extend from psychological assessments of SV to visualization evaluation. Principles and studies in psychology dictate SV relates directly to ability to quickly and accurately compare encodings and layouts of a given visualization along with value estimation, magnitude sense, and subdivisions of charts [58, 92]. Working from these principles, we chose *visual search* as our base paradigm [32], utilizing data from two charts rotated along Cartesian and polar coordinates. We intentionally chose familiar data, chart types, and tasks for novelty and to increase design application across domains. We provide a detailed discussion of study structure and stimuli design below (see supplementary material (SM) for complete stimuli set).

4.1 Recruitment

Participants were recruited through Prolific, a research focused crowdsourcing platform [66]. Prolific was chosen for this study due to its reputation of connecting a pool of diverse and high-quality participants to researchers around the world at fair wages. Additionally, Prolific offers flexible pre-screening, including education and profession which were utilized for this study. The participants were 18 years and older and were fluent in English. Due to the limited number of eligible participants on Prolific when filtered by education and profession, we were able to recruit a balanced sample of 30 participants in each domain with valid data, for a total of 90 participants (details below). Participants were paid £7.71/hour in accordance with Prolific’s fair pay policy. The average response time was 37 minutes and 7s, resulting in an average pay of £4.78 per participant. We collected data across three days and different time slots to diversify the pool of potential participants. We collected data on Jan 28, 2022, Feb 3, 2022, and Feb 17, 2022.

4.2 Apparatus

The online study was created using a Flask Web App with D3.js for chart generation. Participants went through consent and training before each part of the study and were allowed to leave the study at any time,

ending their participation. A trigger warning was presented to participants before agreeing to take part in the study as the data related to the COVID-19 pandemic included case, hospitalization, death, and vaccination statistics. No identifiable data was collected and all data was stored and maintained on a private server at the authors' institution. Flask Web App open source code available at <https://sktandon.github.io/>.

4.3 Study Structure

After consent and demographics collection, the study process consisted of two parts with an additional page of 5-point Likert scale questions, assessing perceived difficulty and personal motivations regarding data visualization (see Section 4.3.6). Participants were asked to confirm readiness to move to the next block between each step to allow for breaks as needed. Training for Part 1 of the study came from the Kit of Factor Referenced Cognitive Tests and included a sample question that participants had to answer correctly to move forward. Active training for Part 2 consisted of 4 sample questions displaying the various charts participants would see throughout the study.

We ran a pilot test to measure the fatigue, length, and complexity of the study. Results from the pilot suggested changes to the study workflow; it became clear that the SV assessment should be placed at the beginning to ensure accurate scores which could otherwise be hindered by fatigue from Part 2, being the longer portion of the study. Additionally, we decided to integrate a break halfway through Part 2, consisting of a catch question wherein users were encouraged to take a break, answer with a specific response, and move on when they were ready. The catch question was not analyzed (other than ensuring all respondents answered correctly for quality assurance) and was not included in response time data analysis.

4.3.1 Screening

Following consent, participants completed a set of 7 demographic questions to collect information about gender, age, education history, profession, and countries of origin and influence. Those who self-identified on Prolific as having studied and working in one of our target areas were admitted and had to reconfirm their primary field of study and work in demographic collection. To reduce confounding factors we only selected participants where education and professional profile were aligned if they were working. No participants whose primary subject and profession failed to align with our targets were included in our analysis. Respondents could only continue through the study if participating on a laptop or desktop computer with a JavaScript enabled web browser to increase consistency of display. Prolific offers a participant score that is based on the quality of an individual's past submissions on their site: our participants had scores of 97% or higher.

We excluded the data of 25 participants from Prolific due to either lack of consistency in primary field of study, data validation failure, or if they experienced a server interruption during the study. We gathered data until we achieved balanced samples with high quality data between our target domains.

4.3.2 Spatial Visualization Assessment

Following demographics, participants began the SV assessment from the Kit of Factor Referenced Cognitive Tests [31], a well-established 2D psychometric assessment [1, 58] largely utilized in previous data visualization studies evaluating SV [50, 57, 65]. This assessment was chosen as SV is elicited in tests involving manipulation and transformation of stimulus [58], which is the primary task in the paper folding test. Additionally, it has the added benefit of a 3-minute time limit for participants to complete the test. This time limit contributes to limiting fatigue of participants taking an online crowdsourced test contributing to higher data quality [11]. We gathered both the selected answer and the response time for each question of the assessment. The test consists of 10 questions in which an image of a paper is folded and punched; participants must then choose from 5 options what the paper will look like when unfolded (see Fig. 2 for a sample question). The SV score is calculated as the number of correct answers out of 10. Congruent to [28, 50, 65, 72] the score was centered by its mean so that participants

above the mean are classified as having high SV, while participants below are classified as having low SV.

4.3.3 Part 2 Stimuli Design

We manipulated three elements to analyze performance over: data density, chart type, and task difficulty.

Data We initially considered synthetic data distributions, however, opted for real-world data to favor familiarity with a data set and scenario applications with participants from a range of domains [11]. The data used in this study was COVID-19 Pandemic data from a comprehensive world statistics site [70]. As data familiarity could increase performance [91], we chose widely consumed data and carefully chose our domains to mitigate effects. The data reflects actual case, vaccination, hospitalization, and death statistics from March 2020–January 2022 across 95 countries. This data was chosen as the scale of the ongoing global pandemic allowed an assumption of a basic level of familiarity with the data and no need for training or expert knowledge [11]. Additionally, the World Health Organization, governments, universities, and data visualization tech companies produced many COVID-19 visual dashboards for international public consumption and policy recommendation during the pandemic – many of these dashboards contained spatially rotated data of varying densities [6, 42, 59, 79, 89].

The data is displayed across two levels of density: 7 and 14 points. A key manipulation for visual search is the number of objects present in the search scene (target(s) + distractor(s) = set size) [32]. We chose 7 as our first set size; 7 is known as a magic number in data visualization, as the limit of the span of absolute judgement and immediate memory sits at about 7 points [60]. This implies that moving sufficiently beyond 7 data points will increase cognitive load; thus we chose 14 as our second set size. Across stimuli, the data was ordered alphabetically by country. Conventionally, visual search studies vary the number of targets and/or distractors present over sub-conditions [32] – we varied the two set sizes amongst the four chart types and three questions detailed below, while maintaining the same density for side-by-side charts.

Chart Types The stimuli draw on elements from both psychology and visualization studies in SV [41, 58, 85]. We formed our stimuli keeping in mind that SV directly affects ability to compare visual encodings and layouts quickly and accurately [58], and that our SV assessment uses 2D rotated stimuli on a Euclidean plane [31]. We chose to display two spatially rotated charts side-by-side to assess performance as participants utilize both charts to respond to a given question. Using two spatially rotated charts side-by-side simulates the cognitive processes involved in SV assessments. The anchoring stimuli chosen for this study was a vertical bar chart. Bar charts are often used as stimuli in cognitive ability and visualization research as they are the most ubiquitous data visualization used over the world [8] and have been shown to capture performance differences between individuals of differing abilities [17, 26]. Additionally, spatially rotated bar charts were heavily utilized in COVID-19 pandemic data dashboards [6, 42, 59, 79, 89]. As points on a Euclidean plane can be displayed using the Cartesian and polar coordinate systems, we use both for our stimuli [46]. We rotated the value-axis of the bar charts in both planes. Anchoring bars on the x-axis in a Cartesian coordinate system provides a vertical bar chart with the value-axis increasing along the y-axis; anchoring bars on the y-axis results in a horizontal bar chart with the value-axis increasing along the x-axis. We then transformed to a polar coordinate plane with value-axis increasing along both (r, θ) . Anchoring bars along a vertical axis with the value scale increasing with θ results in a radial bar chart. Finally, anchoring the bars at the pole with the value-axis increasing with r results in a circular bar plot (see Fig. 1). We followed conventional design principles for circular bar plots, including increasing radial distance on a linear scale given data density, and a large inner radius to reduce perception bias [43]. We note that circular bar plots have some distortion of the bar as the radial distance increases to decrease visual bias toward underestimating small bar heights [12]. Further, previous research demonstrates that although accuracy can be inhibited in polar coordinate charts compared to Cartesian, they are still widely utilized and are aesthetically appealing, thus we used them

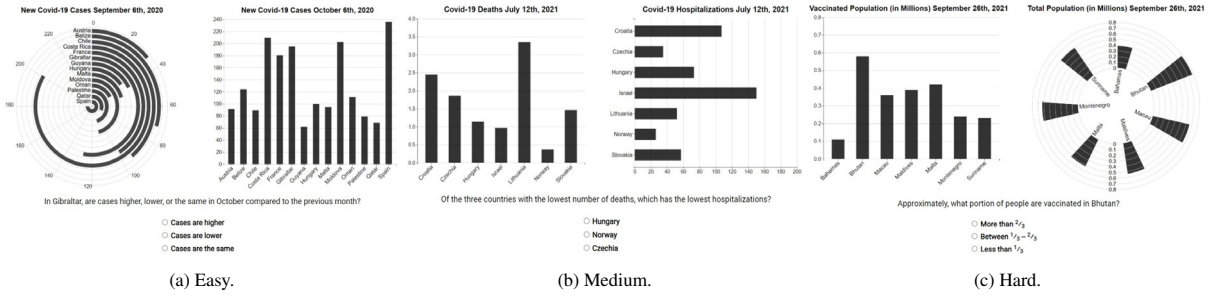


Fig. 3: (a) Example of an Easy question with the Radial chart on left and density 14, (b) Medium question with Horizontal chart on right of density 7, and (c) Hard question with Circular Bar on the right and density 7 (refer to SM for a more comprehensive set).

as sub-conditions in this study [15]. We maintained the vertical bar chart throughout the questions and paired it with either itself or one of the three spatially rotated charts to allow for performance comparison between chart types (e.g., is performance between two Cartesian plots different than a Cartesian & polar chart?). We ensured that the vertical chart appeared on the left and right of the rotated chart across densities and questions. We maintained gray-scale charts to avoid any color interference. As polar coordinate plots require chart lines for readability, we maintained similar lines on the Cartesian charts as well. We varied the four chart types amongst the two set sizes and three questions.

4.3.4 Part 2 Task Design

The basic task paradigm we chose was a visual search process across two charts and densities. Visual search plays an important role in the cognitive process and is a vital element to visualization [10]. As the stimuli consist of two side-by-side charts, the visual search process we chose was a conjunctive search: visual search involving identifying a previously requested target surrounded by distractors possessing no distinct features from the target itself [76]. Conjunctive search involves search over two channels and increases with difficulty as density of distractors increase [76]. We paired conjunctive search with tasks increasing in difficulty (see below) across two levels of density (as described above) to compare performance across varying levels of difficulty. We created three 3-alternative forced-choice multiple-choice questions that required conjunctive use of two chart types for response, we categorized them as Easy, Medium, and Hard. The three response options were randomized across all questions. Questions were randomized after the training section. Fig. 3 shows three examples of different levels of difficulty and density, SM provides a comprehensive overview of all the combinations.

Easy Question The easy question is a search and comparison task. It displays two charts with case numbers from the same set of countries across two months. This question is classified as easy, as it consists of a conjunctive search for the same singular target variable across both displayed charts. Once located, participants compare the target variable across charts, i.e., participants searched for a given country in the first chart and compared case numbers of that country to the second chart to respond if cases were higher, lower, or the same as the previous month. Easy question example: *In Gibraltar, are cases higher, lower, or the same in June compared to the previous month?* Easy Responses: *Cases are higher; cases are lower; cases are the same.*

Medium Question The medium question is a search and comparison task across an increased number of targets. It displays two charts, the left with death rates, the right with hospitalization rates of the same set of countries on a given date. This question is classified as medium as it increases the variables for conjunctive search and comparison to three targets across both charts; participants searched for the three countries with the highest number of deaths in the first chart and responded with which of those countries had the highest hospitalizations from the second chart.

Medium question example: *Of the three countries with the highest number of deaths, which has the highest hospitalizations?* Medium responses consisted of the correct response, one of the countries with the highest number of deaths, but not the highest hospitalizations, and one random country (see Fig. 3).

Hard Question The hard question is a search and comparison task that involves mathematical estimation. It displays two charts, the left with the raw number (in millions) of vaccinated people, the right with the raw population (in millions) of the same set of countries on a given date. This question is classified as hard as after a conjunctive search for one target variable across charts, participants had to perform a mathematical computation to estimate a derived variable. Computing a derived variable is a common task in data analysis and often appears as a sub-task in other operations – the more complex the aggregation the more difficult the interaction [3]. In this question, participants are required to estimate the vaccinated portion of people in a target country. Hard question example: *Approximately what portion of people are vaccinated in Bhutan?* Hard responses: *less than $\frac{1}{3}$, between $\frac{1}{3} - \frac{2}{3}$, more than $\frac{2}{3}$.*

4.3.5 Part 2 Measures

Four possible chart pairings, times two possible layouts (e.g., Vertical or Radial chart on either right or left), times two levels of data density, times three question types/difficulty levels made for a total of 42 multiple-choice questions. To assess performance, we recorded the selected answer and the response time (RT) of each question in all parts of the study. Stimuli were shown randomly to participants to minimize learning effects.

4.3.6 Motivations Assessment

Research shows that personal motivations and cognitive skills (and specifically spatial skills [7, 33]) are closely related and can together explain achievement in STEM subjects [7, 14, 18, 38]. Motivation is part of an individual's goal structures, their belief about what is important, and determines whether they will engage in a given activity [4]; one's educational domain and professional choices influence the activities deemed important [38]. Thus, we chose to evaluate motivations to gain a measurable understanding of how domain background affects interaction with data visualization. We expect domain differences in motivation due to differing experiences and expectations around visualization between disciplines [36, 93, 95]. Additionally, performance on spatial tasks, and tasks involving mathematics, are influenced by self-efficacy, math anxiety, and intrinsic and extrinsic motivators [5, 33, 38]. To increase explainability and psychologically grounded methods in visualization research, we adapted the Science Motivation Questionnaire (SMQ) from [38], which is grounded in prominent theories of academic motivation and encompasses many of the motivational factors found to influence performance in STEM [7]. We chose one question from each of the motivational factors measured by the SMQ: intrinsic, extrinsic, self-efficacy, self-determination, and math anxiety. Each question mirrored the SMQ with the word "science" replaced by "data visualization" and was rated by participants on a 5-point Likert scale as per the questionnaire (see SM).

4.4 Participants

We successfully gathered 30 participants with reliable data in each target domain, giving us a balanced sample, for 90 total participants. We followed a strict rule of balanced samples across groups to support our between subject design. The demographic statistics as a whole and

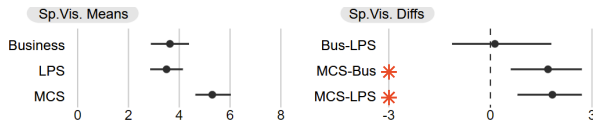


Fig. 4: **H1**: SV per domain, with the CI of means (left) and of mean differences (right). Error bars represent 95% Bootstrap confidence intervals. In the mean differences plot (right), those tighter and farther away from 0 provide stronger evidence of differences. Stars indicate evidence of significant differences across charts.

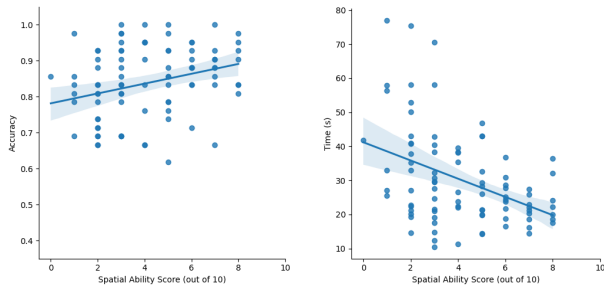


Fig. 5: **H2**: SV score with respect to performance: Pearson correlation between SV and accuracy (left) at $R=0.255$, $p=0.01$. Pearson correlation between SV and response time (right) at $R=-0.408$, $p=0.0006$. Time in seconds.

across the three domains were as follows. Gender participation overall: 51% male, 49% female. Business: 47% male, 53% female. LPS: 30% male, 70% female. MCS: 77% male, 23% female. These are consistent with known educational domain gender differences across Europe [30, 86]. The average age ($\pm$ standard deviation) of all participants was 24 ± 4 , 23 ± 4 for Business, 25 ± 5 for LPS, and 23 ± 4 for MCS. In an effort toward trans-cultural research design in data visualization [2, 9], we opened our study to respondents from all countries. To this end we deployed our study across several days and different times to cater for time zone differences. Overall, we had 49% from the Global South and 51% from the Global North. Business: 57% Global South, 43% Global North. LPS: 57% Global South, 43% Global North. MCS: 33% Global South, 67% Global North. See SM for results related to demographics.

5 RESULTS

We analyzed differences in performance and SV using sample means, hypothesis testing at the $p < 0.05$ level of confidence, and 95% confidence intervals. Confidence intervals (CIs) were constructed in Python using bias-corrected and accelerated bootstrapping (BCa) with 5000 iterations. After testing independent groups for normalcy, we decided to utilize BCa to create confidence intervals along with hypothesis testing using the Mann-Whitney U test to obtain a test statistic and p-value for further validation, as both are robust to non-parametric data [48, 63, 69]. We use both to demonstrate multiple analysis techniques and strength of evidence about the population means, as recommended in recent reviews [29, 39]. Confidence intervals allow for both traditional statistically significant interpretation (if the interval does not overlap 0), as well as subtle differences: the farther from 0 and the tighter it is, the stronger the evidence [80]. Additionally, we analyzed differences in SV using a Pearson's correlation test, where $p < 0.05$ indicates correlation and R ranging 0.1 – 0.3 indicates weak, 0.3 – 0.5 moderate, and 0.5 – 1.0 strong correlation [22]. Below we report on our high-level findings; detailed means and stepwise analysis are reported in SM due to space restrictions.

5.1 H1: Spatial Visualization by Discipline

We hypothesized that SV would differ across discipline with MCS having the highest SV followed by Business, then LPS.

Those in MCS had a mean SV score of 5.3. There is compelling evidence that MCS had higher SV levels than Business ($p < 0.01$) and LPS ($p < 0.001$) by on average 2/10 for both. Business had a mean

score of 3.6 while LPS had a mean score of 3.5 – there was no evidence of difference between them. The CI of mean differences between MCS and Business is (0.6, 2.7) while the CI for MCS and LPS is (0.8, 2.7), indicating a slightly stronger difference between MCS and LPS (see Fig.4). Though Business and LPS have slightly differing spatial abilities according to [87], our finding is in line with [75] which does not show a significant difference between spatial abilities of Business and LPS. Similar to both [75, 87], we found greater mean differences between MCS and Business/LPS than between Business and LPS themselves. See Section 5.4 for further insight.

⇒ We partially confirmed **H1**, SV of participants differed for some domains. The ranking of disciplines in terms of SV is $LPS \approx Business < MCS$. This finding is still consistent with previous research [75, 87] where LPS and Business have similarly low levels of spatial ability, however we found no difference where they find that LPS has slightly lower levels than Business.

5.2 H2: Performance by Spatial Visualization

We hypothesized that performance (accuracy and response time) would differ between those with high and low SV. To determine this, we look at both the correlation between SV and performance, as well as tested sample means and CIs across densities, charts, and questions. For all 90 participants, the mean SV score was 4/10. We had 51 participants with low SV (9 MCS, 22 Business, 20 LPS) and 39 with high SV (21 MCS, 8 Business, 10 LPS). Low spatial individuals had a mean score of 2.5/10, while High had 6/10, meaning high spatial levels were about 4/10 higher (CI(3.3, 4.1), $p < 0.001$).

We found evidence ($p \leq 0.01$) of moderate correlation as SV increases with both accuracy ($R=0.3$) and RT ($R=0.4$) (see Fig. 5). We compared performance of low and high SV participants across the two density levels of 7 and 14 data points, 4 chart types and 3 density levels. We report results from our analysis below (SM reports detailed breakdowns of sample means and CIs).

5.2.1 By Density

Response Time: Mean times were higher for low spatial individuals compared to high spatial for both densities: 8.4s (CI(6, 12), $p < 0.001$) higher for charts with 7 data points and 9.6s (CI(7, 13), $p < 0.001$) higher for charts with 14 data points.

Within low spatial individuals, charts with 14 data points took 4.1s longer than charts with 7 data points (CI(2, 8), $p < 0.001$). High spatial individuals took 2.9s longer on charts with 14 data points (CI(1, 5), $p < 0.001$) compared to charts with 7 data points.

Accuracy: We detected a slightly higher accuracy means (5.2%) on density 7 charts for high spatial individuals than low spatial (CI(0.7, 10), $p < 0.05$). No significant accuracy differences were detected between or within low and high spatial individuals across chart density.

5.2.2 By Chart Type

Response Time: For Vertical charts, low spatial individuals took 6.4s (CI(3, 10), $p < 0.001$) longer than high spatial; for Horizontal low spatial took 9.7s (CI(6, 14), $p < 0.001$) longer; for Radial they took 7.6s (CI(4, 12), $p < 0.001$) longer; and for Circular Bar Plots, low spatial took 11.0s (CI(7, 16), $p < 0.001$) longer.

Within low spatial individuals, polar coordinate charts took 6.0s longer than Cartesian coordinates (CI(2, 10), $p < 0.001$). Within high spatial individuals, polar coordinate charts took 5.2s longer than Cartesian coordinates (CI(4, 7), $p < 0.001$).

Accuracy: There is some evidence that on Vertical charts low spatial means were 5.7% (CI(0.3, 11), $p < 0.05$) lower than high spatial. For Horizontal charts, low spatial means were 5.1% (CI(0.6, 10), $p < 0.05$) lower. On Circular Bar plots low spatial was 7.8% (CI(2, 14), $p < 0.05$) lower. There was no accuracy difference detected for Radial charts.

Within low spatial individuals, polar coordinate chart means were 10.1% lower than Cartesian coordinate charts (CI(5, 15), $p < 0.001$). Within high spatial individuals, polar coordinate chart means were similarly 10.5% lower than Cartesian charts (CI(6, 15), $p < 0.001$).

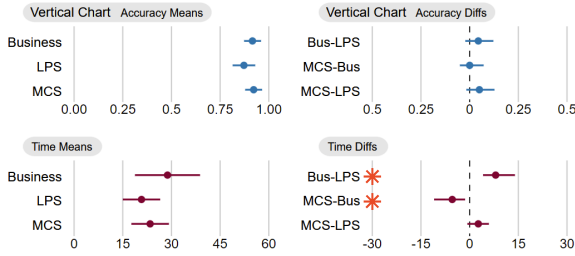


Fig. 6: **H3**: Vertical Bars performance means (left) and mean differences (right), for Accuracy and Response Time. Time in seconds.

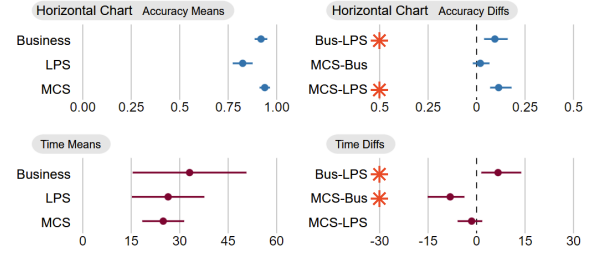


Fig. 7: **H3**: Horizontal Bars performance means (left) and mean differences (right), for Accuracy and Response Time. Time in seconds.

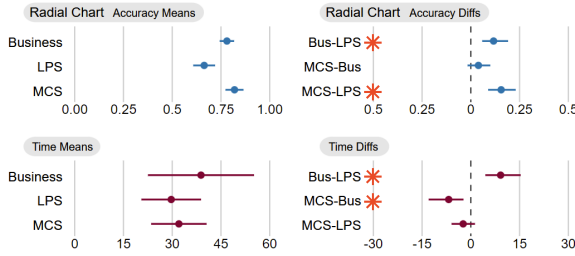


Fig. 8: **H3**: Radial Bar Chart performance means (left) and mean differences (right), for Accuracy and Response Time. Time in seconds.

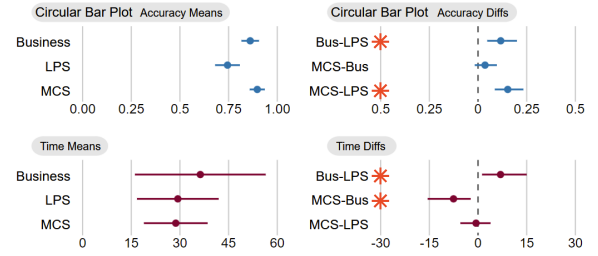


Fig. 9: **H3**: Circular Bar Plot performance means (left) and mean differences (right), for Accuracy and Response Time. Time in seconds.

5.2.3 By Question Difficulty

Response Time: There is compelling evidence that RT means were higher for low spatial individuals across all levels of difficulty: by 8.7s for Easy (CI(6, 12), $p < 0.001$), by 7.5s for Medium (CI(4, 12), $p < 0.01$), and by 10.7s for Hard (CI(7, 15), $p < 0.001$).

Within low spatial individuals, RT means were higher for Medium by 5.7s (CI(2, 10), $p < 0.001$) and Hard by 7.0s (CI(3, 12), $p < 0.001$) when compared to Easy time means, with no significant difference between Medium and Hard questions. For high spatial individuals, time means were also higher for Medium by 7.0s (CI(5, 9), $p < 0.001$) and Hard by 5.1s (CI(3, 7), $p < 0.001$) when compared to Easy times. Interestingly, high spatial time means for Hard questions were 1.9s lower than for Medium times (CI(0.5, 4.3), $p < 0.01$).

Accuracy: Accuracy was similar between low and high spatial individuals in Easy and Medium questions. However, the low spatial means were lower by 10.1% (CI(4, 17), $p = 0.01$) for Hard questions.

Within low spatial individuals, accuracy means were 11.9% lower for Medium (CI(6, 17), $p < 0.001$) and 17.8% lower for Hard (CI(11, 24), $p < 0.001$) when compared to Easy questions. High spatial accuracy means were 16.1% lower for Medium (CI(11, 22), $p < 0.001$) and 12.3% lower for Hard (CI(9, 17), $p < 0.001$) when compared to Easy questions. Neither had notable accuracy differences between Medium and Hard questions within group.

⇒ When looking at overall performance, we confirmed **H2**, SV level does affect RTs (lower SV, longer time to answer questions) and a moderate amount of accuracy. Low and high spatial individuals have a similar discrepancy in performance as density increases and coordinates move between Cartesian and polar.

5.3 H3: Performance by Domain

We hypothesized that performance (accuracy and response time) would differ between disciplines due to both domain background and SV level, with MCS performing best and LPS worst (following their spatial abilities). Looking at overall mean differences, we found that MCS was faster than Business by 7.2s (CI(5, 10), $p < 0.001$), and LPS was faster than Business by 7.6s (CI(5, 11), $p < 0.001$). No difference was detected in overall RT between MCS and LPS. For overall accuracy however, MCS was 12.8% more accurate than LPS (CI(9, 19), $p < 0.001$), and Business was 10% more accurate than LPS (CI(5, 16), $p < 0.001$). No difference was detected in overall accuracy between MCS and Business.

We detail results of our analysis of accuracy and RT across three variables: density, chart type (Figs. 6-9), and question difficulty. Detailed breakdown of the analysis and means are provided in the SM.

5.3.1 By Density

Response Time: For charts of density 7, there is evidence that mean times were higher for Business compared to LPS by 7.6s (CI(4, 13), $p < 0.001$) and MCS by 7.3s (CI(4, 13), $p < 0.001$). There is also evidence mean times were higher for Business in charts of density 14: higher than LPS by 7.7s (CI(4, 13), $p < 0.001$) and MCS by 7.0s (CI(3, 12), $p < 0.001$). There is no evidence of significant differences between MCS and LPS mean times.

Within groups there was similar increased time means for charts of density 14 when compared to density 7: Business by 3.5s, LPS by 3.4s, MCS by 3.8s (all $p < 0.001$).

Accuracy: For charts of density 7, there is strong evidence that MCS had 11.6% higher accuracy (CI(7, 18), $p < 0.001$) and Business had 9.8% higher accuracy means (CI(5, 16), $p < 0.001$) than LPS. Additionally, for charts of density 14, MCS had 14.0% (CI(9, 20), $p < 0.001$) higher accuracy and Business had 10.2% (CI(5, 17), $p < 0.001$) higher accuracy means than LPS. There is no evidence of differences between MCS and Business in accuracy.

No accuracy differences were found between densities within groups.

5.3.2 By Chart Type

See Figs. 6, 7, 8, and 9 for accuracy and RT means and difference CIs.

Response Time: There is evidence that mean times were 8.0s higher for Business than LPS ($p < 0.001$) and 5.4s higher than MCS ($p < 0.05$) for Vertical charts. For Horizontal charts, Business took 6.6s longer than LPS ($p < 0.001$) and 8.2s longer than MCS ($p < 0.001$). Looking at Radial charts, Business took 9.2s longer than LPS ($p < 0.001$) and 6.8s longer than MCS ($p < 0.05$). Last, for Circular Bar Plots, Business took 6.9s longer than LPS ($p < 0.001$) and 7.6s longer than MCS ($p = 0.01$).

Within all domains, there was a similar increase in time between Cartesian and polar coordinate charts: Business by 5.9s (CI(0.5, 11), $p < 0.001$), LPS by 5.0s (CI(2, 8), $p < 0.001$), and MCS by 6.0s (CI(4, 8), $p < 0.001$).

Accuracy: For Vertical charts, there is no evidence of accuracy differences. For Horizontal charts however, there is strong evidence that mean accuracy of MCS was 11.4% higher than LPS and Business

was 9.4% higher (both $p < 0.001$). For Radial charts, there is evidence MCS and Business had higher accuracy means than LPS (by 15.6% and 11.7%, respectively, at $p < 0.001$). For Circular Bars MCS means were higher by 15.3% and Business means higher by 11.7% than LPS, again at $p < 0.001$. There is no significant evidence of difference in means between MCS and Business across chart types.

Within all domains, accuracy means were similarly lower for polar coordinate plots when compared to Cartesian coordinate charts: Business by 9.8% (CI(5, 15), $p < 0.001$), LPS by 13.7% (CI(6, 20), $p < 0.001$), and MCS by 7.5% (CI(3, 12), $p < 0.01$).

5.3.3 By Question Difficulty

Response Time: For Easy questions, there is evidence of increased mean times for Business compared to LPS by 5.4s (CI(2, 10), $p = 0.001$) and MCS by 7.2s (CI(4, 12), $p < 0.01$). Business also has increased times for Medium questions compared to LPS by 9.3s (CI(5, 17), $p < 0.001$) and MCS by 7.3s (CI(3, 15), $p = 0.01$). For Hard questions, Business had increased times by 8.2s compared to LPS (CI(3, 14), $p < 0.001$) and 7.0s to MCS (CI(3, 13), $p < 0.01$). There is no evidence of differences in time means for LPS and MCS.

Within Business there were increased times of 7.6s to Medium questions (CI(2, 15), $p < 0.001$) and 7.0s to Hard questions (CI(2, 13), $p < 0.001$) from Easy questions. For LPS there were increased times of 3.7s to Medium questions (CI(0.6, 7), $p = 0.01$) and 4.3s to Hard questions (CI(0.5, 9), $p < 0.05$) from Easy questions. For MCS there were increased times of 7.6s to Medium questions (CI(5, 11), $p < 0.001$) and 7.2s to Hard questions (CI(5, 10), $p < 0.001$) from Easy questions. No differences were detected in mean times between Medium and Hard questions across domains.

Accuracy: For Easy questions there is some evidence that MCS had higher accuracy than LPS by 9.8% (CI(5, 18), $p < 0.01$) and Business by 3.1% (CI(0.5, 6), $p < 0.05$). Additionally, there was no detected difference between Business and LPS accuracy means for Easy nor Medium questions. There is some evidence that MCS had 11.9% higher scores than LPS in Medium questions (CI(4, 20), $p < 0.01$). For Hard questions, Business had higher accuracy means than LPS by 16.4% (CI(8, 25), $p = 0.01$) and MCS was higher than LPS by 16.7% (CI(8, 25), $p = 0.01$). Other than in Easy questions there was no difference detected between MCS and Business.

For Business, accuracy means fell by 14.2% (CI(10, 20), $p < 0.001$) from Easy to Medium questions and by 11.2% (CI(6, 17), $p < 0.001$) from Easy to Hard. LPS accuracy means fell by 14.5% (CI(5, 23), $p < 0.001$) from Easy to Medium questions and by 20.1% (CI(11, 30), $p < 0.001$) from Easy to Hard. For MCS accuracy means fell by 12.4% (CI(7, 18), $p < 0.001$) from Easy to Medium questions and by 14.0% (CI(10, 19), $p < 0.001$) from Easy to Hard. There was no evidence of differences between Medium and Hard questions within domains.

⇒ Our results partially confirm **H3** (and **H4**). MCS participants tend to have higher accuracy than LPS and lower RTs than Business participants. MCS has higher accuracy than LPS across density, chart types (save Vertical charts), and question difficulty; MCS also has faster times than Business across density, chart types, and question difficulty. However, although SV levels are similar for Business and LPS, Business participants tend to have higher RTs and higher accuracy than LPS. Business and LPS did not differ significantly in accuracy only on Easy & Medium questions, and Vertical charts – although LPS was faster. Overall, we observed a time/error trade-off between Business and LPS – the more time spent, the higher the accuracy. This finding is in line with [41] that found time/error trade offs between domains with similar levels of SV (CS and Education). However, in our study no trade off was detected for MCS participants, who had higher SV levels, and were generally just as fast as LPS and just as accurate as Business. The difference in results could potentially be linked to the fact that we compared across a balanced sample of participants and included education background to professional profile in case profession was misaligned, where [41] did not. This indicates cognitive abilities do interact with domain background to affect performance in both accuracy and RT, and that accuracy can be improved with increased RTs. The latter finding is common in visualization research on SV [28, 41, 82].

Table 1: Motivation scores and significant differences between spatial visualization level and domain. Details in SM.

Motivation	Mean Scores	Significant Differences
Overall (out of 20)	MCS - 12 Business - 12.3 LPS - 9.9	MCS>LPS: CI(0.6, 4), $p < 0.01$ Business>LPS: CI(0.8, 4), $p = 0.01$
Intrinsic (out of 4) Spend my own time learning about data visualization	MCS - 1.8 Business - 1.9 LPS - 1.7	None
Extrinsic (out of 4) My career or studies involve data visualization	MCS - 2.6 Business - 2.4 LPS - 1.6	MCS>LPS: CI(0.6, 2), $p < 0.001$ Business>LPS: CI(0.3, 1), $p < 0.01$
Self-Determination (out of 4) I put effort into learning about data visualization	MCS - 2.2 Business - 3.0 LPS - 2.5	Business>MCS: CI(0.3, 1), $p < 0.01$ Business>LPS: CI(0.03, 1), $p < 0.05$
Self-Efficacy (out of 4) I am confident I will perform well on data visualization tasks	MCS - 2.8 Business - 2.9 LPS - 2.5	Business>LPS: CI(0.03, 0.8), $p < 0.05$
Low Math Anxiety (out of 4) Note: higher score indicates lower anxiety	MCS - 2.6 Business - 2.1 LPS - 1.7	MCS>LPS: CI(0.2, 2), $p = 0.01$

5.4 H4: Motivational Differences

We hypothesized that domain experience would result in motivational differences around interaction with data visualization. As per the SMQ, each motivation response corresponded to a score from 1-4, rating agreement with the statement [38]. The statement measuring anxiety was reverse-scored so that higher scores correspond to lower anxiety – thus, we refer to this construct as low math anxiety. See Table 1 for motivation ratings and significant differences.

The overall average rating of data visualization motivation was 11.4/20. Those above the average motivation score had quicker RTs overall by 7.1s than those below the average (CI(5, 10), $p < 0.001$). They were significantly faster across densities, chart types, and question types (all $p < 0.001$). Additionally, those above the average had higher accuracy in charts of density 14 ($p < 0.05$) and polar coordinate charts ($p < 0.05$) – see SM for details.

Interestingly, both MCS and Business had significantly higher overall motivation scores compared to LPS and had higher accuracy overall. Specifically, Business outranked LPS in extrinsic motivation, self-efficacy, and self-determination (where Business also outranked MCS). MCS outranked LPS only in extrinsic motivation and low math anxiety. This indicates data visualization is more prevalent in MCS and Business domains than LPS and thus there may be a higher value placed on accuracy in visualization tasks. Additionally, putting effort into learning about data visualization and confidence in tasks may increase accuracy – this is aligned with [23, 37] that demonstrate that higher self-determination, self-efficacy, and confidence are correlated with increased performance in mathematics tasks.

⇒ Our results confirm **H4** that domain affects motivations around interacting with data visualization, which in turn affect performance. MCS and Business interact with data visualization and mathematics more often than LPS [36, 93, 95], possibly leading them to have higher motivation and accuracy. Increased accuracy for MCS could be due to lower levels of math anxiety – a major factor in performance of mathematics tasks [14, 37]. High levels of self-determination and self-efficacy could explain why Business out-performed LPS in accuracy across sub-conditions, but had slower RTs. Additionally, high motivation in Business, regardless of SV level, is in line with research demonstrating individuals are often unaware of their cognitive abilities and still choose to interact with and enjoy visual data interfaces i.e., preference to use visualizations is not directly indicative of visual cognitive abilities [54, 88]. These domain differences support ongoing work in information visualization concerning the visualization needs of varying disciplines [27, 53, 62, 68].

6 DISCUSSION

Our results build upon and extend research in information visualization that both SV and domain background influence use of data visualization [53, 83]. These individual differences are often treated independently – we build on work that brings these factors together to increase effectiveness and reach of visualization [41]. We extend research by ensuring globally diverse balanced samples between domains, including education background to professional profile to reduce potential confounding factors, and using prevalent stimuli and tasks. We found differences between domains in statistical performance and motivations around data visualization, leading to design insights and implications.

SV performance We investigated how SV ability and discipline come together to affect task performance (accuracy and RT) on common data visualizations. We confirmed past research that spatial abilities vary with domain [87], with MCS having significantly higher SV than Business and LPS (Fig. 4, **H1**).

Consistent with visualization research into SV [83, 85, 88], we found increased level of SV correlates with higher accuracy and quicker RTs (Fig. 5, **H2**). This finding speaks to space in information visualization for inclusivity, given learning disorders and populations that have been connected to low spatial abilities [13, 35]. RTs differed amongst all sub-conditions (density, chart type, and question difficulty) and accuracy differed for Hard questions amongst all chart types between SV levels. We found a similar discrepancy in performance between Cartesian and polar coordinate charts within low and high spatial individuals. Additionally, we found that for Hard questions (involving math computation) high spatial individuals took less time than with Medium questions, but they maintained significantly higher accuracy than low spatial. This supports research in education psychology that high spatial individuals tend to perform better in mathematics [7, 92].

Domain Performance Between domains, we found behavior consistent with SV level and performance (**H3**) such that MCS had higher accuracy than LPS and quicker RTs than Business. This difference was consistent across sub-conditions. This finding is consistent with research into psychology and visualization that individuals who studied STEM subjects have increased spatial abilities and performance in visualization tasks [7, 83].

Motivation: We found domains had motivational differences around data visualization, affecting their performance (**H4**). We detected an overall time/error trade-off with Business and LPS (i.e., the more time spent on questions the higher the accuracy). We conjecture this trade-off, and difficulty perception, has to do with the motivations and domain differences between Business and LPS. Literature suggests there is higher value placed on data visualization in Business [34, 56, 81, 93] when compared to LPS [45, 95]. This is reflected in the motivations of Business and LPS participants in our study (see Table 1). Notably, MCS and Business reported that data visualization is involved in their career or studies to a higher degree than LPS. Business additionally rated confidence in doing well on visualization tasks as higher than LPS and reported they put more effort into learning about data visualization than both MCS and LPS. These motivations have been shown to correlate with increased performance in mathematics and engagement, performance, and higher quality learning in education [7, 71]. From this, it follows that even with similar levels of SV, those in the Business domain would take time to have higher accuracy on data visualization tasks where those in LPS might not.

The fact that LPS exhibit different abilities and performance should be significant to the visualization community. Individuals in LPS are regularly at the center of government systems and are often counseled by scientific bodies and advisors, especially during the ongoing COVID-19 pandemic [24]. There could be broad reaching consequences if designers fail to cater to the abilities of those making high impact governance decisions when sharing scientific information. Given international & influential bodies like the World Health Organization, governments, universities, and tech companies created COVID-19 dashboards with spatially rotated data [6, 42, 59, 79, 89], it is critical that information visualization is accessible and effective for an imperative group of individuals with huge influence on society.

Design Implications Differences in domain motivations alone have design implications, however these differences combined with SV lead to high impact recommendations for both SV level and specific domains. Across all domains, overall performance is best remaining in Cartesian coordinates and accuracy is unaffected by density, however there are notable differences to account for within each domain. As motivation is high to perform well on data visualization tasks within Business, they can be accurate using rotated visuals: to decrease RT utilizing Cartesian coordinates is recommended. Additionally, Radial charts are the most difficult to process for Business individuals, including Circular Bar Plots. For LPS there is evidence that performance decreases with any visual rotation, thus using one chart type (i.e., Vertical bar) without rotation is recommended. For MCS, performance is high across rotation with smaller differences in performance between Cartesian and polar coordinates. Additionally, research suggests that domain performance on visual tasks is similar between charts regardless of familiarity [27] meaning our recommendations could extrapolate to further chart types.

Outlook & Future Work The scope of this study did not include the causal origins of the different SV abilities amongst disciplines – this is still an open question. It is important to note that other factors can influence task performance such as domain knowledge, representational fluency, visual familiarity, or demographic differences [35, 41, 53]; the interplay between these factors is complex and none alone can explain differences. Regardless of the origins of the differences, our study demonstrates how SV level and domain affect visualization use, which leads to important design implications. Further studies can be done on additional domains to expand knowledge of performance differences in visual tasks and SV between groups for higher impact designs. To increase inclusive design practices, interventions might be studied that allow low spatial individuals to increase accuracy and/or decrease RT needed for visual tasks: benefiting those in LPS and Business domains. As performance drops for LPS and low spatial individuals on visual tasks involving increased math computation, it may be interesting to test if interventions mitigate differences when visual rotation is involved. Our finding that SV level is an important cause behind difference in performance is reflected in intervention research for cognitive abilities [17, 26]. Performance including interventions and guided interactions (e.g., explicit linking across charts, feature highlighting, information redundancy) could be tested and balanced with cognitive load drawbacks for effectiveness.

Our study is the first to combine domain motivations and cognitive abilities of Business, LPS, & MCS to result in novel design implications on common visual stimuli and tasks. This research advances initial work in the visualization community toward cataloguing cognitive differences of domains [41] and is a step toward increasing impact of visual design across disciplines.

7 CONCLUSION

The aim of our research was to build on work in visualization exploring how SV and domain differences come together to affect visualization use. Our study expanded research to the domains of Business and LPS and included real-world data and visual scenarios of spatially rotated data. We provided novel design recommendations based on our findings and presented additional evidence that the interplay between cognitive and demographic factors should be considered to increase effectiveness and inclusivity of visual design. Additionally, our research showed motivational differences between domains that could affect interaction and design needs. As the field of visualization evaluates cognitive and domain differences and the interplay between the two, studies like this can ensure informed and effectual design for all communities.

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